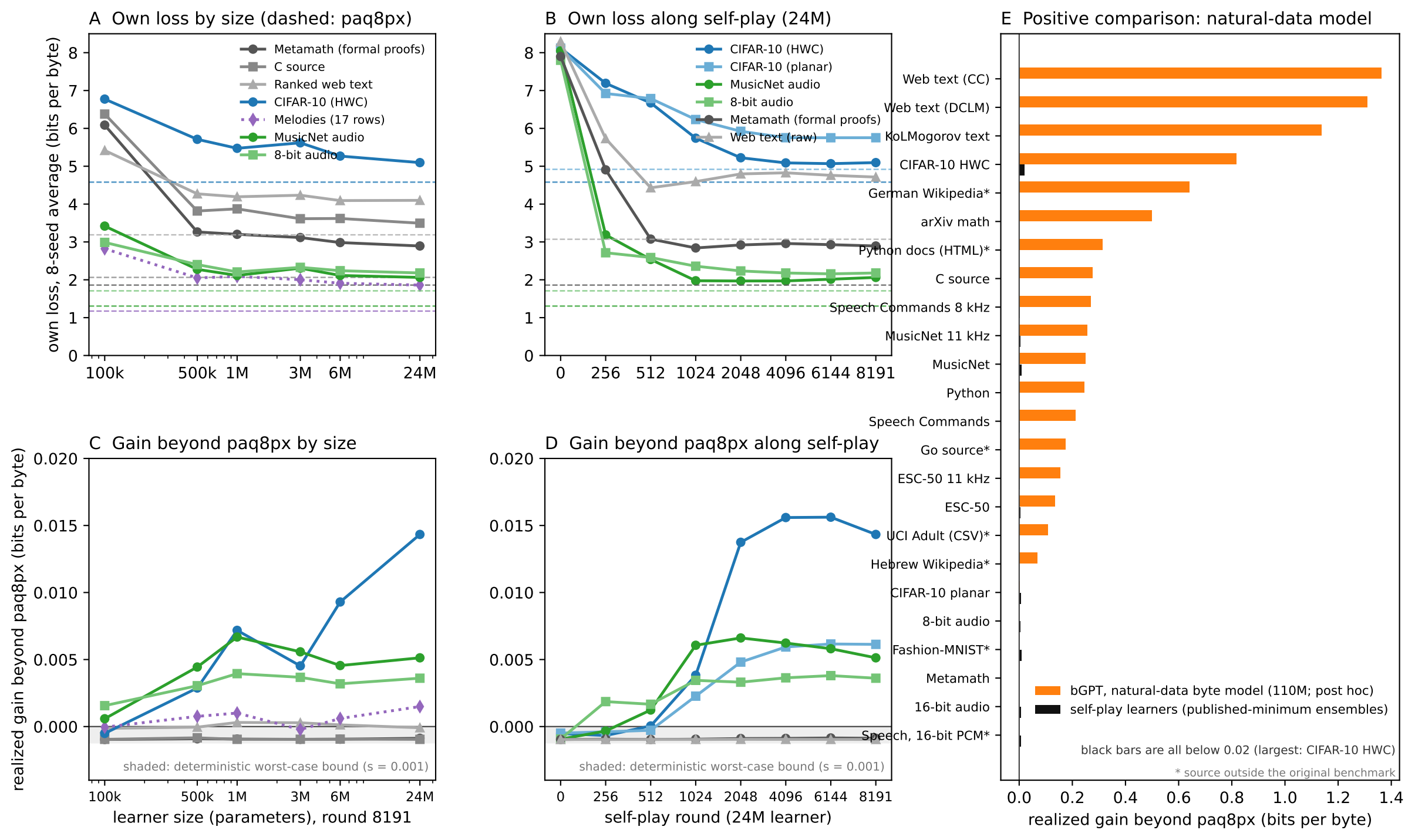
A Own loss by size (dashed: paq8px)
own loss, 8-seed average (bits per byte)
Metamath (formal proofs)
C source
Ranked web text
CIFAR-10 (HWC)
Melodies (17 rows)
MusicNet audio
8-bit audio
100k
500k
1M
3M
6M
24M
B Own loss along self-play (24M)
CIFAR-10 (HWC)
CIFAR-10 (planar)
MusicNet audio
8-bit audio
Metamath (formal proofs)
Web text (raw)
0
256
512
1024
2048
4096
6144
8191
C Gain beyond paq8px by size
realized gain beyond paq8px (bits per byte)
shaded: deterministic worst-case bound (s = 0.001)
learner size (parameters), round 8191
D Gain beyond paq8px along self-play
shaded: deterministic worst-case bound (s = 0.001)
self-play round (24M learner)
E Positive comparison: natural-data model
Web text (CC)
Web text (DCLM)
KoLMogorov text
CIFAR-10 HWC
German Wikipedia*
arXiv math
Python docs (HTML)*
C source
Speech Commands 8 kHz
MusicNet 11 kHz
MusicNet
Python
Speech Commands
Go source*
ESC-50 11 kHz
ESC-50
UCI Adult (CSV)*
Hebrew Wikipedia*
CIFAR-10 planar
8-bit audio
Fashion-MNIST*
Metamath
16-bit audio
Speech, 16-bit PCM*
bGPT, natural-data byte model (110M; post hoc)
self-play learners (published-minimum ensembles)
black bars are all below 0.02 (largest: CIFAR-10 HWC)
* source outside the original benchmark
realized gain beyond paq8px (bits per byte)